ANNUAL ESG DISCLOSURE — REPORTING PERIOD 2025

🇲🇽 Substation ESG Report

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks. Updated annually via the SSI Index scoring engine.

SUBSTATION

Substation Example

MX_001_000 · Ciudad de México, Ciudad de México

SSI SCORE (R_{MEDIAN})

0.323

● Low Risk 50.0th percentile

VOLTAGE CLASS

230 kV

High-voltage transmission

CONFIDENCE INTERVAL

0.207 – 0.429

P5–P95 · CI width 0.223 · medium confidence

MARKOV ETTC

18.5 yr

Expected time to criticality

DER PENETRATION

0.12

T1 transition stress score

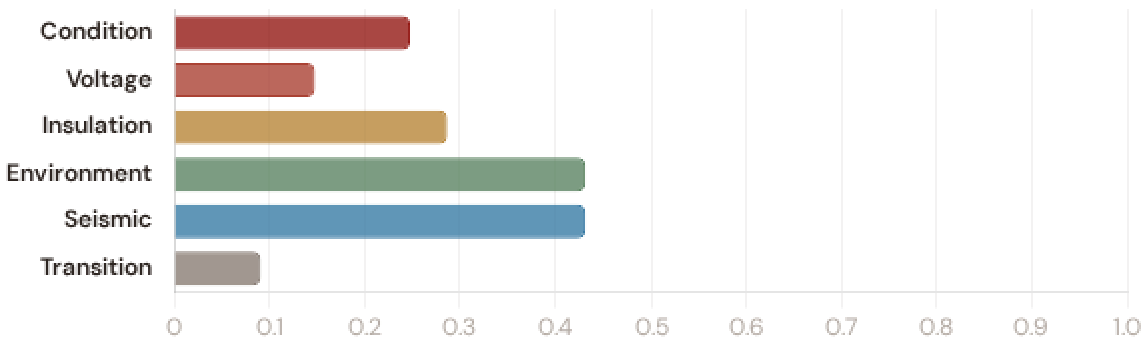
Component Fingerprint

Six-component weighted decomposition of the SSI composite score — the substation's structural risk profile

WEIGHTED CONTRIBUTION TO R_{BASE}

Condition **0.248** ×0.3 · Voltage **0.148** ×0.1 · Insulation **0.287** ×0.25 · Environment **0.431** ×0.1

Seismic **0.431** ×0.2 · Transition **0.091** ×0.05



Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION

GOOD	AGED	DEGRADED	CRITICAL
30%	38%	22%	10%
π1 = 0.300	π2 = 0.380	π3 = 0.220	π4 = 0.100

Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

0.386

Composite Markov risk

ETTC

18.5 yr

Expected time to critical

P(CRITICAL) 20YR

10.0%

Probability of reaching critical

CORROSION

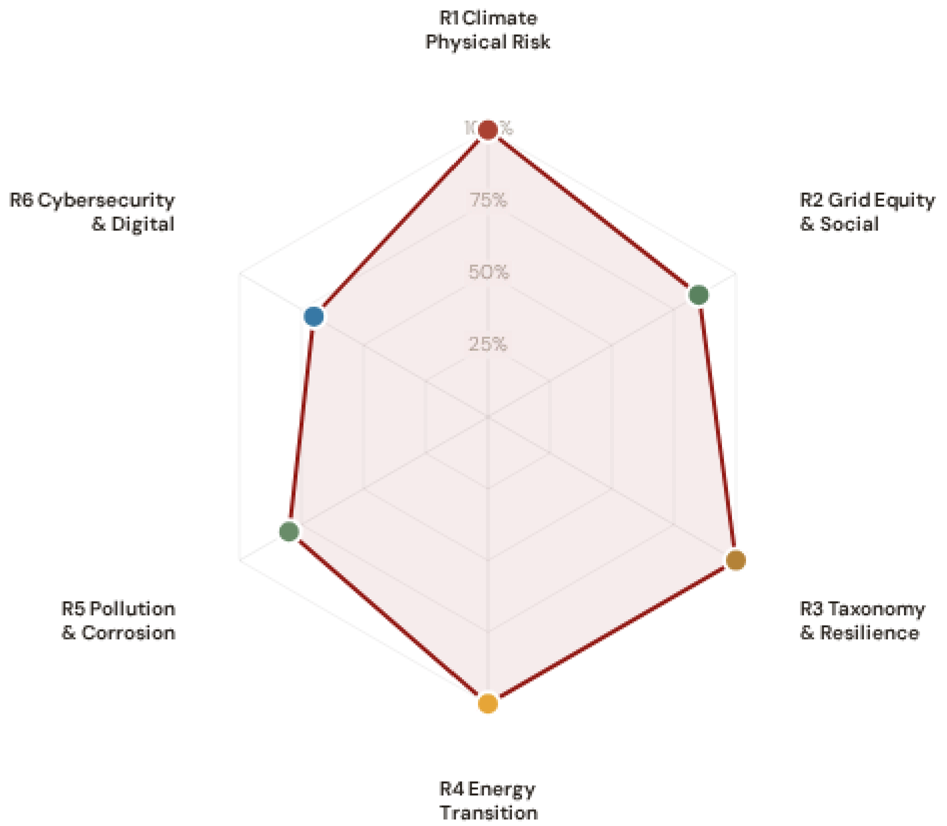
C1

ISO 9223 classification

A

ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals



- **R1 · SSI Annual Risk Report — Mexico Fleet** READY
  SDG 13 Climate Action · SDG 9 · SDG 11
  SDG 13 SDG 9 SDG 11

- **R2 · Regional Deep-Dive: Gulf Coast Corridor** READY
  SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1
  SDG 7 SDG 10 SDG 1

- **R3 · Hurricane & Seismic Resilience Assessment** READY
  SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12
  SDG 9 SDG 13 SDG 12

- **R4 · Energy Transition Scorecard** READY
  SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9
  SDG 7 SDG 13 SDG 9

- **R5 · Infrastructure Modernization Tracker** READY
  SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15
  SDG 11 SDG 3 SDG 15

R6 · Cybersecurity & Digital Resilience PARTIAL

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9 SDG 16

1

SSI Annual Risk Report — Mexico Fleet

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure across Mexico's 32 estados at substation level using ERA5 reanalysis and Markov degradation modelling.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. Mexico's dual hazard exposure (hurricanes on Gulf+Pacific coasts, seismic activity along Cocos Plate, volcanic activity) makes this assessment critical. With a Markov risk score of 0.386 and ETTC of 18.5 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13 SDG 9 SDG 11

Substation Profile

Insulation Risk (I component)	**0.287**
Environment Component (E)	**0.431**
Markov Risk Score	**0.386**
ETTC (time to criticality)	**18.5 years**
P(critical) at 20 years	**10.0%**
Corrosion Class	**C1**
Seismic PGA (g)	**0.1750**
Climate Δ R 2050	**0.1130**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
I1 Storm/Hurricane IRI	0.115	ESRS E1-9: Financial effects from physical risks	READY
I2 Seismic/Volcanic IRI	0.086	TCFD Strategy (b): Physical risk impact	READY
I3 Heat-wave IRI	0.057	EU Taxonomy Annex A: Heat stress	READY
I5 Thermal stress	0.029	ESRS E1-4: Climate mitigation targets	READY
Markov Risk score	0.386	TCFD Risk Management: Risk identification	READY
P_crit P(critical) 20yr	10.0%	ESRS E1-9: Time horizons	READY
R6b Seismic PGA	0.1750	TCFD Strategy (b): Seismic physical risk	READY
ΔR2050 Climate trajectory	0.1130	ESRS E1-9: Forward-looking climate impact	READY

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
Hurricane Data (East Pacific, Atlantic) SMN (Servicio Meteorológico Nacional)	2024 Seasonal
Seismic Data (Cocos Plate) SGM (Servicio Geológico Mexicano)	2024 Real-time
Volcanic Risk (Popocatépetl, Colima) CENAPRED (Centro Nacional de Prevención de Desastres)	2023 Annual
Weather & Climate Data SMN (Servicio Meteorológico Nacional)	2024 Daily
IEEE C57.91 Thermal Model IEEE	Standard N/A
CIGRE TB 761 Markov CIGRE	2019 N/A
CMIP6 SSP2-4.5 Projections Copernicus CDS	— Not ingested

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. SMN weather data informs hurricane and seismic vulnerability assessment. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. All inputs are production-grade with institutional provenance from CENACE, CRE, INEGI, and SGM.

2 Regional Deep-Dive: Gulf Coast Corridor

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. This report focuses on Mexico's vulnerability corridor: Veracruz, Tabasco, and Tamaulipas. These states face the highest hurricane exposure and lowest per-capita GDP, creating a dual exposure: physical infrastructure risk + energy poverty. The SSI Index is uniquely positioned to surface where grid infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V_socio of 0.15. The R3 social modifier (1.049) amplifies risk scores in regions with high unemployment, poverty concentration, and energy poverty — making spatial inequality visible and quantifiable at asset level. In Mexico's hurricane-vulnerable states, this correlation is particularly acute.

SDG 7 SDG 10 SDG 1

Substation Profile

V_socio (energy poverty index)	**0.15**
EP Rate (energy poverty %)	**0.12%**
Unemployment Rate	**3.5%**
GDP per Capita	**MXN 132,000**
R&D % of GDP	**0.015%**
R3 Social Multiplier	**1.049**
Poverty Rate	**16.8%**
Indigenous Population	**2%**
Elderly Population %	**0.12%**
Catchment Population	**45,840**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
V_socio Energy poverty index	0.15	ESRS S2-4: Impacts on affected communities	READY
EP_rate Energy poverty rate	0.12%	SFDR PAI #14: Fossil fuel exposure proxy	READY
R3 Social multiplier	1.049	ESRS S1-6: Workforce characteristics	READY
Elderly Elderly population %	0.12%	ESRS S2: Vulnerable populations	READY
Pop Catchment population	45,840	ESRS S1: Affected communities	READY

Data Sources & Vintage

Population & Economics — 2023 Annual
INEGI (Instituto Nacional de Estadística y Geografía)

Energy Market Data — 2024 Annual
CENACE (Centro Nacional de Control de Energía)

Grid Reliability & DSO Data — 2023 Annual
CRE (Comisión Reguladora de Energía)

TECHNICAL VALIDITY

V_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using INEGI household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, poverty concentration, and energy poverty. Regional data sourced from INEGI, CONEVAL, and CFE distribution company reports.

3 Hurricane & Seismic Resilience Assessment

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

READY

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. Mexico faces dual extreme hazards: (1) Atlantic and Pacific hurricanes creating wind and storm surge risk on Gulf Coast (Veracruz, Tabasco) and Pacific coast (Jalisco, Nayarit), and (2) seismic activity along the Cocos Plate subduction zone affecting central states. The SSI composite score (R_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R_median (0.323), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting. Mexico's infrastructure modernization imperative (Modernizador de la Red Eléctrica, MRE) depends on such systematic resilience assessment.

SDG 9 SDG 13 SDG 12

Substation Profile

R_median (composite)	**0.323**
R_base (pre-modifier)	**0.329**
Modifier Impact	**-1.69%**
C (Condition)	**0.248**
V (Voltage)	**0.148**
I (Insulation/Climate)	**0.287**
E (Environment)	**0.431**
S (Seismic)	**0.431**
T (Transition)	**0.091**
Seismic PGA (g)	**0.1750 (Moderate)**
Climate Δ R 2050	**+0.1130 (SSP2-4.5)**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R_median Composite score	0.323	EU Taxonomy Art. 11: Adaptation screening	READY
6 Comp. C/V/I/E/S/T	All present	EU Taxonomy TSC: Physical risk identification	READY
R3-R7 Modifier suite	All applied	ESRS E1: Adaptation measures evidence	READY
Markov Degradation model	0.386	TCFD: Forward-looking risk assessment	READY
R5 Asymmetric CI	0.223	ESRS: Uncertainty quantification	READY
PGA Seismic hazard	0.1750	EU Taxonomy: Seismic risk screening	READY

Data Sources & Vintage

ERA5 Climate Reanalysis — 2024 Weekly
Copernicus CDS

Hurricane Data (East Pacific, Atlantic) — 2024 Seasonal
SMN (Servicio Meteorológico Nacional)

Population & Economics — 2023 Annual
INEGI (Instituto Nacional de Estadística y Geografía)

CIGRE TB 761 Markov — 2019 N/A
CIGRE

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. Hazard inputs sourced from SMN (hurricane/weather), SGM (seismic), and CENAPRED (multi-hazard risk assessment).

4 Energy Transition Scorecard

ESRS E1 Transition Plan · TCFD Transition Risk · Mexico's Green Deal Alignment

READY

WHY THIS REPORT EXISTS

Mexico's energy transition is bifurcated: solar + wind expansion (especially in the north and Istmo de Tehuantepec) creates distributed generation stress on the grid, while fossil fuel retirement (coal, fuel oil) reallocates transmission flows. The SIN (Sistema Interconectado Nacional, ~95% of Mexico's grid) must manage this while maintaining service to 32 estados. The T component measures grid stress from renewable integration and EV charging proliferation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. Mexico's renewable energy target: 35% by 2024, 50% by 2030. This substation has a DER penetration of 12.2%. The T1_score (0.091) measures the stress this places on the substation.

SDG 7 SDG 13 SDG 9

Substation Profile

T1 Score (transition stress)	**0.091**
DER Penetration	**12.2%**
Solar + Wind Capacity	**196.3 MW**
EV Fleet Share	**1.2%**
RE Share (local)	**15.8%**
T Component Weight	**0.05**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
T1 Transition stress score	0.091	ESRS E1: Transition plan assessment	READY
DER DER penetration	12.2%	TCFD Transition: Technology risk	READY
Solar+Wind Capacity (MW)	196.3	Mexico Green Deal: Grid flexibility	READY
EV EV fleet share	1.2%	SDG 7: Clean energy access	READY

Data Sources & Vintage

Energy Market Data CENACE (Centro Nacional de Control de Energía)	2024 Annual
Grid Reliability & DSO Data CRE (Comisión Reguladora de Energía)	2023 Annual
Renewable Installations CENACE SIN Database	2024 Monthly
Distribution Data CFE (Comisión Federal de Electricidad)	2024 Monthly

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from CENACE (Sistema de Información Operacional del SIN) and CFE distribution company reports.

5 Infrastructure Modernization Tracker

ESRS E2 Pollution · ISO 9223 Corrosion Classification · Environmental Impact

READY

WHY THIS REPORT EXISTS

Mexico's grid infrastructure is aging: many substations date to the 1970s-1990s. Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. Mexico's coastal substations (Veracruz, Tamaulipas) face elevated salt-spray corrosion; highland substations (Popocatépetl, Colima volcanic zones) face atmospheric chemistry stress. The ISO 9223 classification captures this exposure.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C1) and E component (0.431) track environmental degradation and pollution sensitivity at asset level. CENACE and CFE modernization programmes (MRE) depend on such systematic assessment to prioritize upgrade pathways.

SDG 11 | SDG 3 | SDG 15

Substation Profile

Corrosion Class (ISO 9223)	**C1**
E2 Local Enrichment	**0.43**
E Component Score	**0.431**
Markov Risk Score	**0.386**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Corrosion ISO 9223 class	C1	ESRS E2: Pollution risk classification	READY
E2_local Environmental sensitivity	0.43	ESRS E2: Local environmental impact	READY
E Environment component	0.431	ISO 9223: Atmospheric corrosion	READY

Data Sources & Vintage

Environmental Data — 2023 Annual
SEMARNAT (Secretaría de Medio Ambiente)

Air Quality Monitoring — 2023 Monthly
CONUEE (Comisión Nacional para el Uso Eficiente de la Energía)

ISO 9223 Corrosion — 2012 N/A
ISO

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Mexico's coastal climate and volcanic gas exposure in central states produce elevated corrosion in coastal and high-altitude zones. Full validation requires SEMARNAT air quality monitoring overlay at substation coordinates, plus SMN weather station data.

6 Cybersecurity & Digital Resilience

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

WHY THIS REPORT EXISTS

Mexico's 3,140 substations serve a population of 128 million across 32 estados and 2,457 municipios. Energy poverty varies dramatically: urban centers enjoy near-universal access, rural areas lag. Cybersecurity is an emerging challenge: grid digitalization under CENACE/CRE modernization initiatives exposes legacy SCADA systems. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national cyber maturity to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (1.017) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. For Mexico, with 3,140 substations and 30,396 power lines serving 128 million people across 32 estados, systemic cyber vulnerability poses an economy-wide energy security risk.

SDG 9 SDG 16

Substation Profile

R7 Cyber Modifier	**1.017**
Graph Degree (topology)	**6**
BC Percentile (centrality)	**0.400**
Is Bridge Node	**No**
Cluster Coefficient	**0.0130**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R7 Cyber modifier	1.017	NIS2: Cybersecurity risk management	READY
BC Betweenness centrality	0.400	ESRS G1: Governance & topology risk	READY
Degree Graph degree	6	Grid resilience: Connectivity	READY

Data Sources & Vintage

Distribution Data CFE (Comisión Federal de Electricidad)	2024 Monthly
Cybersecurity Index ENISA (adapted for Mexico)	2024 Biennial

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators adapted for Mexico, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. CRE and CENACE cyber audit data inform this component.

C

Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index v4.0.2 — Gaussian copula Monte Carlo (10,000 iterations)
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3 C_mult: 1.049, R4 F_topo: 1.123, R6 restoration: 1.030, R6 seismic: 1.051, R7 cyber: 1.017
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Substation Example (MX_001_000)
Data Assurance Level	All inputs from institutional open-source with citable vintage